DAR-Dub: A Multi-Agent Director-Actor-Reviewer Framework for Scenario-level Emotionally Consistent Movie Dubbing

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A Dataset

To overcome the limitations of isolated utterance-level datasets and enable scenario-level iterative modeling, we introduce the DAR-Animation Dataset. By expanding the original V2C-Animation benchmark to include 100 movies, we construct a corpus of approximately 120 hours containing 109,632 utterances organized into about 2,400 scenarios. Following rigorous data partitioning principles, the dataset is randomly split at the scene level into 60% training, 10% validation, and 30% test subsets.

To ensure a vast spectrum of emotional diversity, inter-character interaction dynamics, and complex scenario-level dramatic transitions, the 100 animated films were carefully curated from industry-leading studios, including Disney, Pixar, DreamWorks, Illumination, Sony Pictures Animation, and Blue Sky Studios. All video materials were legally acquired through authorized digital distribution platforms, primarily Amazon Video and Disney+, ensuring high-fidelity audiovisual quality for scenario-level modeling. The complete list of the 100 movies is as follows: *The Boss Baby*, *The Boss Baby 2*, *Brave*, *Cloudy with a Chance of Meatballs*, *Cloudy with a Chance of Meatballs 2*, *Coco*, *The Croods*, *The Croods 2*, *How to Train Your Dragon*, *How to Train Your Dragon 2*, *How to Train Your Dragon 3*, *Frozen*, *Frozen II*, *The Incredibles*, *Incredibles 2*, *Inside Out*, *Inside Out 2*, *Meet the Robinsons*, *Moana*, *Wreck-It Ralph*, *Ralph Breaks the Internet*, *Tangled*, *Tinker Bell*, *Tinker Bell and the Lost Treasure*, *Tinker Bell and the Great Fairy Rescue*, *Toy Story*, *Toy Story 2*, *Toy Story 3*, *Toy Story 4*, *Up*, *Zootopia*, *Abominable*, *Sing*, *Sing 2*, *Despicable Me*, *Despicable Me 2*, *Despicable Me 3*, *Despicable Me 4*, *The Secret Life of Pets*, *The Secret Life of Pets 2*, *Antz*, *Bee Movie*, *Big Hero 6*, *Bolt*, *A Bug's Life*, *Captain Underpants: The First Epic Movie*, *Cars*, *Cars 2*, *Cars 3*, *Chicken Little*, *Elemental*, *Encanto*, *Finding Dory*, *Finding Nemo*, *Flushed Away*, *Home*, *Kung Fu Panda*, *Kung Fu Panda 2*, *Kung Fu Panda 3*, *Kung Fu Panda 4*, *Lightyear*, *Luca*, *Madagascar*, *Madagascar: Escape 2 Africa*, *Madagascar 3: Europe's Most Wanted*, *Penguins of Madagascar*, *Megamind*, *Monsters Inc*, *Monsters University*, *Monsters vs. Aliens*, *Mr. Peabody & Sherman*, *Onward*, *Over the Hedge*, *Puss in Boots*, *Puss in Boots: The Last Wish*, *Ratatouille*, *Raya and the Last Dragon*, *Rise of the Guardians*, *Shark Tale*, *Shrek*, *Shrek 2*, *Shrek the Third*, *Shrek Forever After*, *Soul*, *The Bad Guys*, *The Good Dinosaur*, *Trolls*, *Trolls World Tour*, *Turbo*, *Turning Red*, *Minions*, *Minions: The Rise of Gru*, *The Mitchells vs. the Machines*, *Wish*, *Strange World*, *Migration*, *Spies in Disguise*, *Ferdinand*, *Epic*, *Smallfoot*.

A.1 Open-Source Declaration and Ethical Disclaimer

The dataset is constructed strictly for non-commercial academic research. To uphold intellectual property rights while ensuring full scientific reproducibility, we strictly refrain from distributing raw copyrighted video files. Instead, we provide the research community

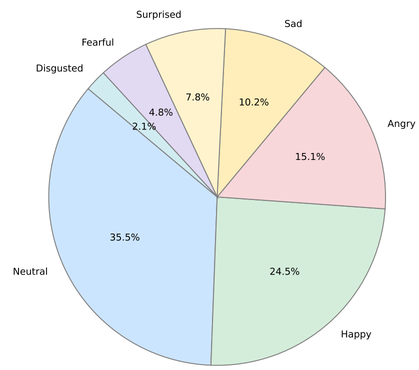

Figure 1: Proportional distribution of the seven categorical emotion labels across the DAR-Animation dataset.

with a comprehensive suite of pre-extracted multimodal features and automated construction scripts to facilitate future academic inquiry. Researchers can utilize these provided scripts to automatically process their locally obtained movie copies into the exact scenario-level formats required by our framework. The dataset construction tools, pre-extracted features, code, and interactive demos are made publicly available at our repository.

A.2 Distribution of Emotion Labels

To provide a comprehensive understanding of the affective diversity within the DAR-Animation dataset, we detail the distribution of emotional labels across the corpus. Recognizing that human emotion in cinematic scenarios is highly complex, we provide multi-dimensional emotional annotations for every utterance. This includes discrete categorical labels and continuous Valence, Arousal, and Dominance (VAD) values extracted via MERALION-SER, capturing both macro-level emotional states and subtle, fine-grained affective shifts.

Figure 1 illustrates the proportional distribution of the seven primary categorical emotions (Neutral, Happy, Angry, Sad, Surprised, Fearful, and Disgusted). The distribution inherently reflects the natural ebb and flow of narrative-driven animated films. While "Neutral" dialogue appropriately constitutes the majority of the corpus, providing a baseline for conversational grounding, the dataset maintains a robust representation of high-intensity emotional peaks essential for evaluating dynamic scenario-level movie dubbing. While categorical labels offer an intuitive macro-level classification, they often fall short in representing the nuanced intensity and intra-class variance of professional voice acting. To address this, Figure 2 utilizes a grouped violin plot to map the continuous VAD dimensions—Valence (negative to positive), Arousal (calm to

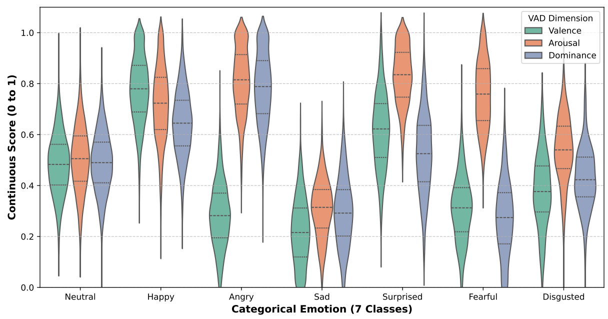

Figure 2: The distribution of continuous Valence, Arousal, and Dominance (VAD) scores across the seven categorical emotions.

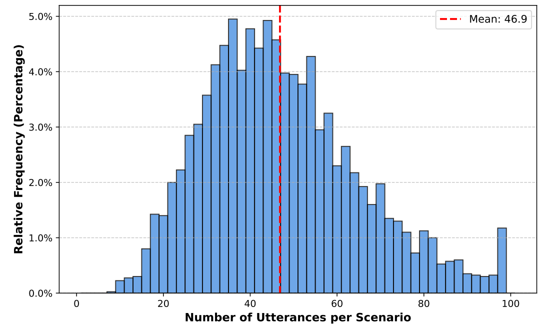

Figure 3: Histogram detailing the distribution of utterance counts per scenario.

active), and Dominance (weak to strong), all normalized to a $[0, 1]$ scale—across the seven categorical emotions.

This dimensional analysis serves to explicitly validate the internal consistency and high annotation quality of the DAR-Animation dataset. As demonstrated in Figure 2, the continuous distributions strictly align with established psychological models:

- **Happy** utterances are tightly clustered at high Valence and elevated Arousal.
- **Angry** features predominantly low Valence, but sharply high Arousal and Dominance, accurately capturing aggressive tension.
- **Sad** is characterized by corresponding drops in both Valence and Arousal, reflecting a low-energy, negative affective state.

Crucially, the varying widths of the violins visualize the substantial intra-class variance within our dataset. Rather than mapping discrete labels to static acoustic templates, this distribution proves that our dataset captures a continuous spectrum of varying intensities, ranging from mild annoyance to explosive rage within the "Angry" category. This fine-grained affective coverage is instrumental for our multi-agent framework, as it forces the model to learn subtle scenario-level emotional transitions rather than memorizing stereotypical tones.

A.3 Example of Scenario-level Organization

Rather than arbitrarily slicing the movie into fixed-length chunks or static physical locations, our segmentation strategy is strictly event-centric. A scenario is defined as a continuous block of time in which a single, cohesive narrative event unfolds. The segmentation adheres to the following core principles:

Event-Centric Coherence: A scenario must encapsulate a complete narrative beat or conversational event, such as an escalating argument, a continuous chase sequence, or an ongoing investigation task. Crucially, a scenario is not strictly bound by static spatial limits; if an event moves across different physical locations, such as characters walking from a building to the street while arguing, it remains a single scenario. This ensures the model learns the complete emotional arc of the interaction.

Temporal Continuity: While spatial boundaries can be crossed, the scenario must represent an unbroken, continuous timeline. Any transition that involves a significant temporal leap, including a fade-to-black, a "next day" cut, or a time-skip montage, immediately breaks the emotional inertia and thus triggers a scenario boundary. This guarantees that intra-scenario emotional transitions remain physically and logically plausible.

To quantitatively validate this segmentation strategy and our core motivation, Figure 3 illustrates the relative distribution of utterance counts per scenario across the DAR-Animation dataset. The histogram demonstrates a pronounced prevalence of multi-turn interactions(averaging over 45 utterances per scenario) with the vast majority encompassing sustained, continuous dialogues rather than isolated remarks.

To demonstrate the integration of our structured event-based data into the Multi-Agent framework, Table 1 presents a representative scenario instance (*Scene: Zootopia018*) showcasing our multi-layered annotation schema. Designed for high-fidelity movie dubbing, this schema establishes situational awareness at the Global Level through comprehensive *Scene Descriptions* and detailed *Character Personas*. At the Utterance Level, we introduce the *Director Arc* to provide nuanced stylistic and performance guidance, which is further enriched by fine-grained affective metrics—including categorical *Emotion* labels and continuous *VAD* (Valence, Arousal, Dominance) values—alongside precise *Duration* constraints to ensure cross-modal alignment.

B Supplement Experiments

B.1 Efficiency and Computational Cost

To evaluate the computational footprint of our framework, we employ ProDubber as the primary baseline. Given that our Actor agent inherits ProDubber's acoustic architecture, this comparison isolates the incremental costs introduced by the Director's strategic guidance and the Reviewer's iterative feedback. Table 2 summarizes the VRAM usage, token consumption, and Real-Time Factor (RTF).

VRAM and Token Consumption. The token count reflects the average "cognitive overhead" per scenario. For a representative scene comprising approximately 47 utterances, the *Ours (w/o Reviewer)* configuration consumes ~3.9k tokens, primarily dedicated to the Director's initial scenario-level analysis. In the full pipeline, this increases to ~8.8k tokens, as the Director must process the

Table 1: An Exemplary Instance (*Scene: Zootopia018*) of Scene-level Annotation

Scenario Context & Character Personas						
Scene Desc.	Judy and Nick confront the mastermind, Bellwether, in a museum. The duo orchestrates a brilliant ruse... baiting a confession.					
Judy	A brilliant and resilient rabbit officer who uses her vulnerability to orchestrate a final sting operation...					
Nick	A master con artist turned hero; his acting skills are crucial as he fakes 'going savage' to trick Bellwether...					
Bellwether	The true mastermind; her voice shifts from sweet and timid to cold, manipulative, and megalomaniacal...					

Structured Utterance-Level Annotation Workflow

Idx	Char.	Transcript	Director Arc (Instructional Guide)	Emotion	VAD (V, A, D)	Dur.
17	Bellwether	"Come on out, Judy."	Smooth, resonant, and radiating an effortless arrogance.	Happy	(0.53, 0.69, 0.66)	2.00s
18	Judy	"Take the case."	Strained, low-volume, and vibrating with authentic pain.	Fearful	(0.34, 0.57, 0.58)	1.24s
19	Judy	"Get it to Bogo."	Hushed, urgent, and saturated with a final professional plea.	Fearful	(0.35, 0.58, 0.59)	1.21s
20	Nick	"I'm not gonna leave you behind..."	Steady, mid-energy, and radiating unshakable loyalty.	Angry	(0.32, 0.65, 0.65)	1.60s
21	Judy	"I can't walk."	Broken, trembling, and leaking physical-vulnerability grief.	Fearful	(0.22, 0.68, 0.67)	1.37s
22	Nick	"We'll think of something."	Soft, firm, and shimmering with protective cleverness.	Fearful	(0.38, 0.64, 0.64)	1.00s

Table 2: Comparison of Efficiency and Computational Overhead. (RTF: Real-Time Factor, lower is faster. Actor RTF refers to the acoustic synthesis stage, while Total RTF encompasses LLM interaction and iterative generation latency)

Methods	VRAM (GB)	Avg. Tokens	Actor RTF (↓)	Total RTF (↓)	Avg. Iters (↓)
ProDubber	1.08	N/A	0.18	0.18	1.00 (Fixed)
Ours (w/o Reviewer)	2.98	~3.9k	0.22	0.55	1.00 (Fixed)
Ours	21.38	~8.8k	0.22	1.02	1.14

Reviewer's textual critiques as additional context. Regarding hardware requirements, the transition to the full framework increases VRAM usage to 21.38 GB. This jump is primarily attributed to the local deployment of the 7B-parameter Reviewer Agent (operating at FP16 precision) to minimize evaluation latency, while the Director Agent is accessed via a remote API.

Inference Latency and Iteration Dynamics. The core acoustic synthesis (Actor RTF) remains highly efficient, with a marginal increase from 0.18 to 0.22 caused strictly by the EmotionGateformer. System-level latency (Total RTF) scales with LLM reasoning and the feedback loop: the Director-only mode reaches 0.55, while the full pipeline averages 1.02. Crucially, an average iteration rate of 1.14 indicates that the Director-Actor tandem successfully passes our rigorous quality threshold (> 4.5) on the first attempt for 86% of utterances. The Reviewer intervenes—triggering higher token usage and latency—only for the remaining 14% of emotionally complex or prosodically demanding cases. Considering that high-fidelity movie dubbing is fundamentally an offline industrial task, we argue that trading near-real-time latency ($RTF \approx 1.0$) for substantial gains in scenario-level emotional consistency is a highly favorable trade-off for professional production.

B.2 Analysis of Foundation Model Efficiency

To investigate the impact of model scale on the quality of dubbing guidance, we conducted a scalability study. To rigorously isolate the foundational reasoning capabilities from the compensatory effects of the Reviewer's feedback loop, all evaluations in this section were performed using the non-iterative *Ours w/o Reviewer* configuration. As demonstrated in Table 3, we observed a significant performance jump when transitioning from 7B to 30B, with SEACCC and SECT improving by 15.0% and 43.8% respectively. Notably, this leap is achieved with only a marginal increase of approximately 26% in average token consumption. Furthermore, when deploying the heavyweight Qwen3-VL-235B, the performance gain is minimal, yielding an improvement of merely 0.008 in SEACCC, despite the average token cost surging by more than 3.0 times. This empirical evidence suggests that our DAR-Dub framework, specifically the Director Arcs, effectively saturates the reasoning requirements of the task at the 30B scale. The Qwen3-Omni-30B model thus serves as the optimal foundation, providing a sweet spot that delivers emotional alignment with superior industrial-grade efficiency.

Table 3: Comparison of different foundation models as the Director Agent

Model	Params	SEACCC (↑)	SECT (↓)	Avg. Tokens
Qwen2.5-Omni	7B	0.621	0.048	~3.1k
Qwen3-VL	235B	0.722	0.024	~11.8k
Ours (Qwen3-Omni)	30B	0.714	0.027	~3.9k

B.3 Impact of Top-K Retrieved References

To investigate the influence of retrieval pool size on the Actor agent's performance, we varied the number of retrieved reference acoustic emotion vectors K from 1 to 10. To strictly isolate the impact of the retrieval mechanism from the compensatory effects of the iterative feedback loop, all evaluations in this section were conducted using the non-iterative *Ours w/o Reviewer* configuration.

As illustrated in Figure 4, the model's performance, characterized by EMO-ACC and SEACCC, initially improves as K increases, reaching its peak at $K = 2$. This indicates that a parsimonious set of highly relevant references effectively provides the necessary acoustic emotional priors for the Actor agent. However, as K exceeds 3, a

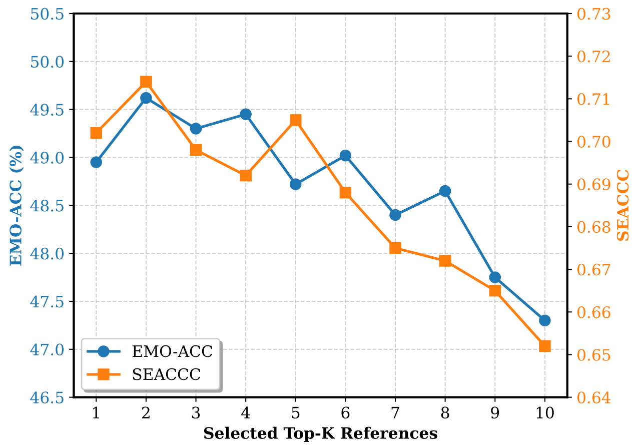

Figure 4: Analysis of retrieval parameter.

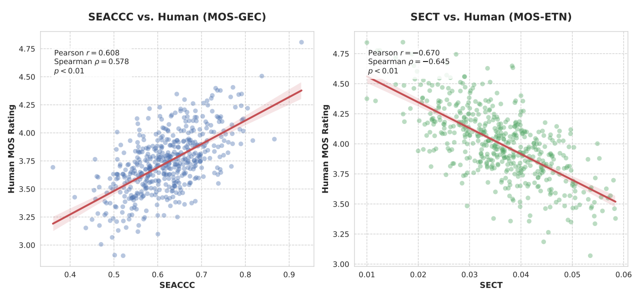

Figure 5: Correlation between SEACCC and human MOS-GEC (left) and SECT and MOS-ETN (right).

gradual performance degradation is observed. This decline suggests that retrieving excessive samples introduces redundant information, which dilutes the precision of the scenario-level guidance provided by the Director Arc. Consequently, we set the default retrieval size to $K = 2$ to strike an optimal balance between affective clarity and computational efficiency.

B.4 Objective Metric Alignment Analysis

To validate the effectiveness of our proposed objective metrics in mirroring human perception, we conduct a correlation analysis between the objective scores (SEACCC and SECT) and the subjective MOS ratings. This analysis is performed on the same set of approximately 500 utterances used in the subjective test. Specifically, we align the calculated SEACCC with the corresponding human MOS-GEC ratings to evaluate global emotional consistency, and align SECT with MOS-ETN ratings to assess emotional transition naturalness.

As illustrated in Figure 5, SEACCC achieves a Pearson $r = 0.608$ (Spearman $\rho = 0.578$) with $p < 0.01$, demonstrating a strong positive correlation with human judgment of global affective arcs. Furthermore, SECT exhibits a significant negative correlation with MOS-ETN, reaching $r = -0.670$ (Spearman $\rho = -0.645$) with $p < 0.01$. Given that SECT is defined as the Mean Squared Error (MSE) of emotional trajectories, this negative correlation explicitly confirms that a reduction in objective emotional deviation directly translates to an improvement in perceived transition naturalness. These high correlation coefficients and significance levels verify that SEACCC and SECT serve as reliable automated proxies for human-centric quality standards in scenario-level movie dubbing.

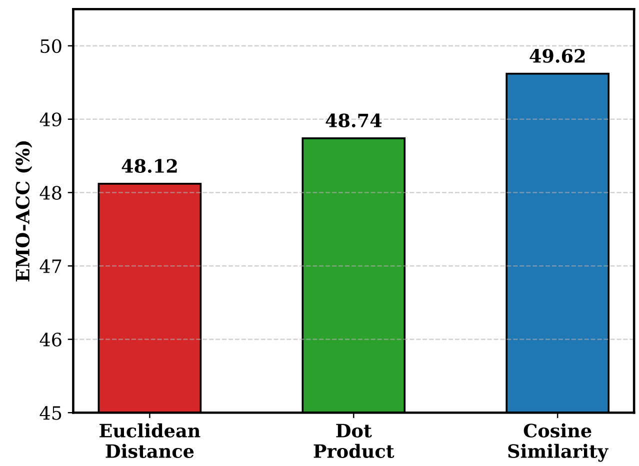

Figure 6: Comparison of similarity metrics for Director Arc matching.

B.5 Analysis of Retrieval Similarity Metrics

To determine the most effective distance metric for the RAG mechanism, we compare Cosine Similarity, Dot Product, and Euclidean Distance for matching the current Director guidance vector D_t with the historical database D_{base}. Consistent with the Top-K analysis, this evaluation is also conducted on the non-iterative base model to strictly assess the retrieval quality. As illustrated by the bar chart in Figure 6, Cosine Similarity clearly outperforms the other two metrics, achieving the highest EMO-ACC of 49.62%. Specifically, it provides a noticeable performance gain over Dot Product at 48.74% and Euclidean Distance at 48.12%. This superiority stems from the nature of Director Arc embeddings D_t which encapsulate high-level semantic and emotional directions; thus, Cosine Similarity effectively captures the angular alignment of these affective tones. These findings validate our choice of Cosine Similarity as the optimal metric for experience-guided retrieval in movie dubbing tasks.